

Desiderata

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We are looking for a computable $\mathbf{A}$ (Alpha, algorithm, agent) that behaves as follows when facing the below Ω (environment, problem, Omega).

Definition 1. A *decision tree* $\bar{T}$ is a rooted countable tree (WLOG of degree two) such that

- Every node is annotated as a decision by either $\mathbf{A}$ or Ω .
- Each next node is computed by an algorithm Round taking as input all past rounds and $\mathbf{A}$'s and Ω 's behavior in them.
- Each Ω -decision is taken in one of the following three ways:
 - *Empirical coin*: The probability of heads is annotated in the node (although WLOG all coins are 0.5). It's a random bit completely unpredictable to $\mathbf{A}$ or Ω .
 - *Logical coin*: A decidable sentence is annotated in the node. Ω goes left iff the sentence is true. Analogously we can use halting computations instead of decidable sentences.
 - *Entanglement*: Ω uses a certain logical inductor $\mathbb{P}$ over the deductive process $\bar{D}$, where D_n equals the union of $\{\text{Round}(\bar{o}_{<n}) = \text{Round}(\bar{o}_{<n})\}$ with either $\{\mathbf{A}(\bar{o}_{<n}) = \mathbf{A}(\bar{o}_{<n})\}$ (if last node was an $\mathbf{A}$ -decision) or $\{\Omega(\bar{o}_{<n}) = \Omega(\bar{o}_{<n})\}$ (if it was an Ω -decision, either a logical coin or an entanglement). Ω calculates $\mathbb{P}_i((\neg)\mathbf{A}(\bar{o}) = a \mid \varphi)$ for a certain observation history $\bar{o}$, action a , logical sentence φ and $i \in \mathbb{N}$ smaller or equal to the present round number. Ω 's decision is a computable function of this value. Both an algorithm for this function and the expression $\mathbb{P}_i((\neg)\mathbf{A}(\bar{o}) = a \mid \varphi)$ are annotated in the node.
- Every node is labeled with a (finite) utility U , that $\mathbf{A}$ obtains when that node is reached.
- Also, we probably want $\mathbf{A}$ to be able to check the whole tree at any time (although it will only be able to check a finite portion each time).
- Entanglements between $\mathbf{A}$ -decisions?

Definition 2. Given an action a in a decision tree $\bar{T}$, $\text{Future}(a)$ is the sub-tree after the edge a . We sometimes abuse notation and talk about the Future of a finite recursively chosen action sequence, instead of a single action.

Definition 3. A decision tree $\bar{T}$ satisfies the *coordination-free assumption* if, whenever i and j are incomparable $\mathbf{A}$ -decision nodes (and $\bar{o}_i, \bar{o}_j$ observation histories ending at those nodes), and $a_i^1, a_i^2, a_j^1, a_j^2$ different actions available at i and j respectively with $\text{Future}(a_i^1) \neq \text{Future}(a_i^2)$, then none of the two include an entanglement of the form $\mathbb{P}_n((\neg)\mathbf{A}(\bar{o}_j) = a_j^l \mid \varphi)$.

It's a bit too restrictive to ask for identity between Futures, since two Futures could possibly be non-identical and yet trivially equivalent. But this is no problem, because in that case we'll just be able to rewrite the tree such that they really are identical.

Also, we don't actually care about any incomparable $\mathbf{A}$ -decision nodes, just those such that $\mathbf{A}$ and $\mathbb{P}_n$ can't confidently enough decide $\mathbf{A}(\bar{o}_j) = a_j^l$ from $\bar{o}_i$. But what we mean by "confidently enough" here is fuzzy (depending on complexities, and possibly changing online), so again it's more straightforward to syntactically cut these cases out.

Definition 4. A decision tree $\bar{T}$ satisfies the *catastrophe-free assumption* if

1. For any node i , there is an infinite branch above i whose accumulated utility per node dominates that of any other branch in the whole tree.
2. For each $\bar{o}$ and a in the tree, there are only finitely many entanglements of the form $\mathbb{P}_i((\neg)\mathbf{A}(\bar{o}) = a \mid \varphi)$.

That is, it is always possible for $\mathbf{A}$ to learn the right behavior and obtain the highest possible reward in the limit (since there's always a path for that, and Ω doesn't infinitely punish a single action).

This is satisfied, for example, in infinite sequences of finite decision trees without cross-entanglements (defined below). The assumption is also satisfied provided (2), and that the decision problems presented by Ω don't depend on $\mathbf{A}$'s actions (maybe I should define this formally, calling it a *fixed* tree).

Desiderata 1 (LIDT rewrites to UDT). *Say $\mathbf{L}$ is the LIDT algorithm (with arbitrary initial state, and hard-coded ε -exploration) and faces a decision tree $\bar{T}$ satisfying the coordination-free assumption and the catastrophe-free assumption. And additionally there are infinitely many nodes i with the following property:*

- *There is a (single) $a_i \in \mathcal{A}_i$ and $k \in \mathbb{N}$ such that after a_i there are k forced actions $a_{i+1} \dots, a_{i+k}$, and $\text{Future}(a_{i+1}, \dots, a_{i+k}) = \text{Future}(\mathbf{A}(\bar{o}_i, \bar{T}_{i \leq j \leq i+k}, \bar{T}))$,¹ and the node arrived at by a_j has utility U exactly that obtained by $\mathbf{A}$ in that round. Also, no entanglement anywhere in the tree refers to the a_i action in particular (for example arbitrarily punishing it), only to the actions in the sequence.*

That is, there always are infinitely many opportunities to rewrite to UDT for a finite amount of time. Then, for late enough i , $\mathbf{L}$ chooses such an a_i , or at least there's no other action it strictly prefers.

Also, if there are such nodes with arbitrarily large k , then for late enough i , if $a, a' \in \mathcal{A}_n$ are such actions with $k < k'$, $\mathbf{L}$ chooses a' , or at least doesn't strictly prefer a .

Sketch of Proof. For each action, $\mathbf{A}$ considers $\mathbb{E}_n(U_{>i} \mid \mathbf{A}(\bar{o}_i) = a)$, where $U_{>i} := \delta \cdot U_{i+1} + \delta^2 \cdot U_{i+2} + \dots$ is time-penalized. $\mathbf{A}$ thus learns to reason about what comes after its action, including taking into account arbitrary logical considerations about how $\mathbf{A}$ itself will take the following actions (and also how $\mathbf{L}$ will take them, once it gains control back). This "learning about its future decisions (which in turn include learning about how these decisions affect future repetitions, etc.)" is our

¹The definition of how $\mathbf{A}$'s actions are calculated is straight-forward: $\mathbf{A}$ experiences a different decision tree, in which it only has control of actions during rewrite, and $\mathbf{L}$'s decisions are certain logical coins.

slow way of obtaining "converges on the optimal global policy for a good finite number of the next steps", for a weak definition of optimal (and requiring the coordination-free assumption). So, in a sense, for late enough i , $\mathbf{A}$ will already be "very good" at reaping most of the benefit of the next few rounds (hopefully as "good" as an algorithm can get). Thanks to ε -exploration, $\mathbf{L}$ will try this out often enough, and learn how well $\mathbf{A}$ performs.

One worry: What if our decision tree includes arbitrary adversarial elements towards updateless agents, or towards rewriting? Well, we probably want to add as an assumption that the a_i are not directly punished, that is, not entanglement cares about them. This captures "caring only about the actions played, not about the rewrite". But is there some other fuzzier way to obtain this bad behavior, like "checking $\mathbf{A}$'s code for whether it's an updateless agent"? Apparently not, since all Ω can check are actions, and it cannot use a finite-time action to always check which algorithm you were (before you had time to reason about which algorithm you want to be), because of the catastrophe-free assumption. $\square$

It's worth noting that $\mathbf{A}$ sees the whole tree, and knows it will go back to being $\mathbf{L}$. Thanks to $\mathbf{L}$ caring about future time-penalized utilities (just like $\mathbf{A}$), in the limit $\mathbf{A}$ will be able to enact real optimality, instead of just trying to reap fast rewards so that $\mathbf{L}$ self-modifies more in the future.

Desiderata 2 (Tiling result). *Given a decision tree $\bar{T}$ satisfying the coordination-free assumption and the catastrophe-free assumption, suppose for some node i we have:*

- *There is an $a_i \in \mathcal{A}_i$, $k \in \mathbb{N}$, and a fixed policy $\bar{b}$ for the next k rounds (a function from each $\mathbf{A}$ -decision to actions). After a_i there are k forced actions $a_{i+1}, \dots, a_{i+k}$, with $\text{Future}(\bar{a}) = \text{Future}(\bar{b})$, and the node arrived at by a_j has utility U exactly that obtained by the fixed policy $\bar{b}$ in that round. And entanglements are resolved in the obvious way, by just checking the counterfactual value of $\bar{b}$.*

That is, a node where $\mathbf{A}$ can precommit to follow a finite policy. Then, for late enough i , $\mathbf{A}$ doesn't strictly prefer a to any other action.

Sketch of Proof. As above, it learns about its actions and expected rewards, and future instances of itself, etc. So it will just learn to take the action that makes it act optimally for the next few rounds, and it will also learn that committing to the best actions is no different from taking them. $\square$

There's a general theme here: Our way to obtain "logical-time-0 single-agent omniscience without dynamic inconsistency" is by having the agent learn more and more about versions of itself and their behavior, thus climbing towards complete self-understanding, and pushing the dynamic instabilities further and further away.

Desiderata 3 (Replicates LIDT when no entanglements). *Given a decision tree $\bar{T}$ satisfying the catastrophic-free assumption, assume further that, if $\bar{o}_i$ is not an extension of $\bar{o}_j$, then in no Ω -decision in or after i do we have an entanglement of the form $\mathbb{P}_i((\neg)\mathbf{A}(\bar{o}_j) = a \mid \varphi)$. Then, for late enough i , if $\bar{o}_{\mathbf{A}}$ and $\bar{o}_{\mathbf{L}}$ are the observation histories up to i of $\mathbf{A}$ and $\mathbf{L}$ respectively (suppose they both exist, that is, both reach that node), we have $\mathbf{A}(\bar{o}_{\mathbf{A}}) = \mathbf{L}(\bar{o}_{\mathbf{L}})$.*

Sketch of Proof. Notice the new assumption implies coordination-free.

We want to see that, when their LIs are advanced enough, they agree on the conditional assessments of action-values. This will be because, given the simplicity of the future decision tree, even $\mathbf{L}$'s traders naively maximizing for their selfish benefit will provide the "global optimum for a few

rounds" that **A** approximates. In fact, probably they both will easily converge on "the actual global optimum" in this case. A way to prove this could be: in this case, because of the structure of the decision tree, any optimal policy can be reduced to "picking a sequence of optimal myopic actions within our counterfactual". **A**'s LI learns that this is the case (and doesn't need to worry about entanglements), and **L**'s LI doesn't even consider this, and just learns to correctly assess the myopic options.

One worry: Because of small contingencies in their past histories, both LIs have pretty different observations, and forever disagree action-relevantly on some logical coins. We probably just need to include another assumption here.

One observation: Maybe we can also prove something in terms of average accumulated reward, even if they don't literally reach the same node. $\square$

Definition 5 (Logical muggings). Consider a sequence of decidable logical sentences $\bar{\phi}$, generated by a certain algorithm. Consider the decided version of this sequence

$$(\neg)\phi_n := \begin{cases} \phi_n, & \text{if } \phi_n \text{ is true} \\ \neg\phi_n, & \text{if } \phi_n \text{ is false} \end{cases}$$

Say $\mathbb{P}$ is a logical inductor over the latter. Assume that

$$\lim_{n \rightarrow \infty} \min\{\mathbb{P}_n(\phi_n), 1 - \mathbb{P}_n(\phi_n)\} \neq 0$$

(if the above limit exists). That is, $\mathbb{P}$ doesn't learn to decide the sequence in a timely manner.

We will define a decision tree $\bar{T}$.

In each round n , **A** is shown the sentence ϕ_n along with its truth value. Then

- If ϕ_n , **A** can choose to Pay 100\$, or Not Pay them.
- If $\neg\phi_n$, **A** automatically receives $\mathbb{P}_n(\text{"A}(\bar{o}_{<n}, \text{Round}(\bar{o}_{<n})) = \text{Pay} \mid \phi_n) \cdot 1.000\$$.

Desiderata 4 (Solves Logical muggings). **A**'s accumulated reward per round in Logical muggings dominates those of a certain class of algorithms, for each sequence $\bar{\phi}$ in a certain class. That is, there is always a later round where **A** has better accumulated reward per round.

Sketch of Proof. Probably the class of the $\bar{\phi}$ s can be as big as "computable sequences" (if Ω knows their generating program). But note that, even if $\bar{\phi}$ is easy enough for **A** to compute, we want **A** to act updateless.

It would seem like the class of algorithms could also be all algorithms, since regardless of your complexity (whether you can decide the coins), you can just be updateless about them. But this might not be as straight-forward: it is conceivable that complex **As** can exploit certain simple Ω s (by "defecting on a non-e.c. sub-sequence"), and I should try to find an example of this.

Now on to the actual proof:

If $\bar{\phi}$ were complex enough for $\mathbb{P}_n(\phi_n)$ to not be decided, but syntactically simple enough for it to Learn all relevant Pseudorandom Patterns (as well as **A**'s complexity can), we could just use that. We might be able to adapt this to other situations, by choosing slower or faster LIs (although I think there are cases where no sweetspot exists). In any event, this all feels misguided, because we're talking about "actual limited beliefs / knowledge", instead of having the correct knowledge but using it in the globally optimal way. Main proposals to explore here are Omni-traders / BLI (through Coherence Constraints) and Vetoing traders. $\square$

Definition 6 (Parfit’s Hitchhiker). Two different rounds alternate infinitely:

1. In round n even, Ω calculates $\mathbb{P}_n(\mathbf{A}(\bar{o}_{<n+1}) = \text{Pay} \mid \mathbf{\Omega}(\bar{o}_{<n}) = \text{Save})$ and plays Save iff it’s $> p$. (This action-relevant discreteness is the main difference.)
2. In round n odd, if Ω played Save in the previous round $\mathbf{A}$ receives 1.000\$, and then chooses whether to Pay 100\$. If Ω played Not save, nothing happens.

In a call, we thought this posed a fix-point problem (that needed to be solved by ε -exploration), because for advanced enough states

$$\mathbb{P}_n(\mathbf{A}(\bar{o}_{<n+1}) = \text{Pay} \mid \mathbf{\Omega}(\bar{o}_{<n}) = \text{Save}) = \mathbb{P}_n(\mathbf{A}(\bar{o}_{<n+1}) = \text{Pay} \mid S_n > p) =: S_n$$

I think we were wrong, and this depends more finely on $\mathbf{A}$. If $\mathbf{A}$ does something very simple, like actually Paying a fraction s of the time (by consulting a hard Pseudorandom Sequence), then some trader in $\mathbb{P}$ will arbitrage this, and this belief will be held correct (even when $\mathbb{P}_n(S_n > p)$ limits to 0, if $s \leq p$).

Desiderata 5 (Solves Parfit’s Hitchhiker).

Definition 7 (Logical muggings with Math Homework). Same as Logical muggings, but to pay up $\mathbf{A}$ must tell Ω the right π digit (choice between 10 digits).

Desiderata 6 (Solves Logical muggings with Math Homework).

Definition 8 (Infinite sequence of finite trees: Non-delegated version). Any decision tree $\bar{T}$ with the following form: $\mathbf{A}$ is presented (no matter what its actions) an infinite sequence of finite decision trees, and each node only has entanglements within its finite tree.

Desiderata 7 (Solves Finite non-delegated trees). *Given an infinite sequence of finite trees $\bar{T}$ satisfying the coordination-free assumption, our algorithm dominates.*

Sketch of Proof. The catastrophe-free assumption follows.

If we fully solved the above Desiderata about particular cases, probably this follows easily through combinations of the same mechanism (and explicit search over policies). $\square$

Definition 9 (Infinite sequence of finite trees: Delegated version). In each individual round, $\mathbf{A}$ is shown a whole tree, decides on a deterministic policy $\bar{b}$ (a function from each $\mathbf{A}$ -decision to actions, respecting entanglements between $\mathbf{A}$ -decisions), and precommits to it. (So counterfactual entanglements are trivially defined.)

Desiderata 8 (Solves Finite delegated trees).

If Desiderata 2 (Tiling) is satisfied, it would seem like our agent will eventually take the same actions in the Delegated and Non-delegated versions. But maybe there’s some relevant difference because of using these trivially defined counterfactuals instead of LI counterfactuals?

On another note, when in an updateful setting we’ll have $\mathbf{A}$ satisfy nice LI properties because of acting like LIDT. But maybe we can get some of them even in updateless settings.